$$\frac{\Gamma \vdash e_1 : \tau \to \mathsf{bool} \rightarrowtail e_3 \qquad \Gamma \vdash e_2 : \tau \rightarrowtail e_4}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) : \tau \rightarrowtail \mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_3, e_4\rangle\rangle}\ \text{Compile-Flat}$$

$$\frac{\Gamma \vdash e_1 : \tau_2 \rightarrowtail e_2}{\Gamma \vdash \lambda f : \tau_1 \to \tau_2.\lambda x : \tau_1.\mathsf{mon}_j^{k,l}\Big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail e_3}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail e_3\ e_2}\ \text{Compile-Func}$$

$$\frac{\begin{array}{c} f, x \notin \mathrm{dom}(\Gamma) \cup \{y\} \cup \mathrm{fv}(\kappa_1, \kappa_2) \qquad \Gamma \vdash e_1 : \tau_1 \to \tau_2 \rightarrowtail e_2 \\ e_w = \lambda f : \tau_1 \to \tau_2.\lambda x : \tau_1.\mathsf{mon}_j^{k,l}\Big(\kappa_2\Big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\Big], f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\ \Gamma \vdash e_w : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail e_3 \end{array}}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail e_3\ e_2}\ \text{Compile-Dep}$$

$$\frac{\begin{array}{c} x \notin \mathrm{dom}(\Gamma) \qquad \Gamma, x : \langle \tau_1, \tau_2\rangle \vdash \mathsf{mon}_j^{k,l}(\kappa_1, \mathsf{inj.l}(x)) : \tau_1 \rightarrowtail e_1 \\ \Gamma, x : \langle \tau_1, \tau_2\rangle \vdash \mathsf{mon}_j^{k,l}(\kappa_2, \mathsf{inj.r}(x)) : \tau_2 \rightarrowtail e_2 \qquad \Gamma \vdash e : \langle \tau_1, \tau_2\rangle \rightarrowtail e_3 \end{array}}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\langle \kappa_1, \kappa_2\rangle, e) : \langle \tau_1, \tau_2\rangle \rightarrowtail (\lambda x.\langle e_1, e_2\rangle)\ e_3}\ \text{Compile-Tuple}$$

$$\mathtt{h}_{\mathscr{C}} : \mathtt{lab} \quad = \quad \{\mathscr{C} \to \forall\alpha.\langle \mathtt{blab}, \langle \alpha \to \mathsf{bool}, \alpha\rangle\rangle \to \alpha\}$$

$$\mathsf{wrap} \quad = \quad \mu w : \forall\alpha.\big(\big(() \to^{\langle \mathtt{h}_{\mathscr{C}}\rangle} \alpha\big) \to \alpha\big).\Lambda\alpha.\ \lambda f : \big(() \to^{\langle \mathtt{h}_{\mathscr{C}}\rangle} \alpha\big).$$

$$\mathtt{handler}_d(\{\mathscr{C} \to$$

$$\lambda\langle l, \langle e, v\rangle\rangle.\lambda k.\ \text{if } w[\mathsf{bool}]\ (\lambda_.e\ v) \text{ then } k\ v \text{ else } \mathtt{error}_l\})\ f$$

where $\lambda\langle l, \langle e, v\rangle\rangle$ deconstructs the argument as tuple pattern

Figure 1: Compiler Rules